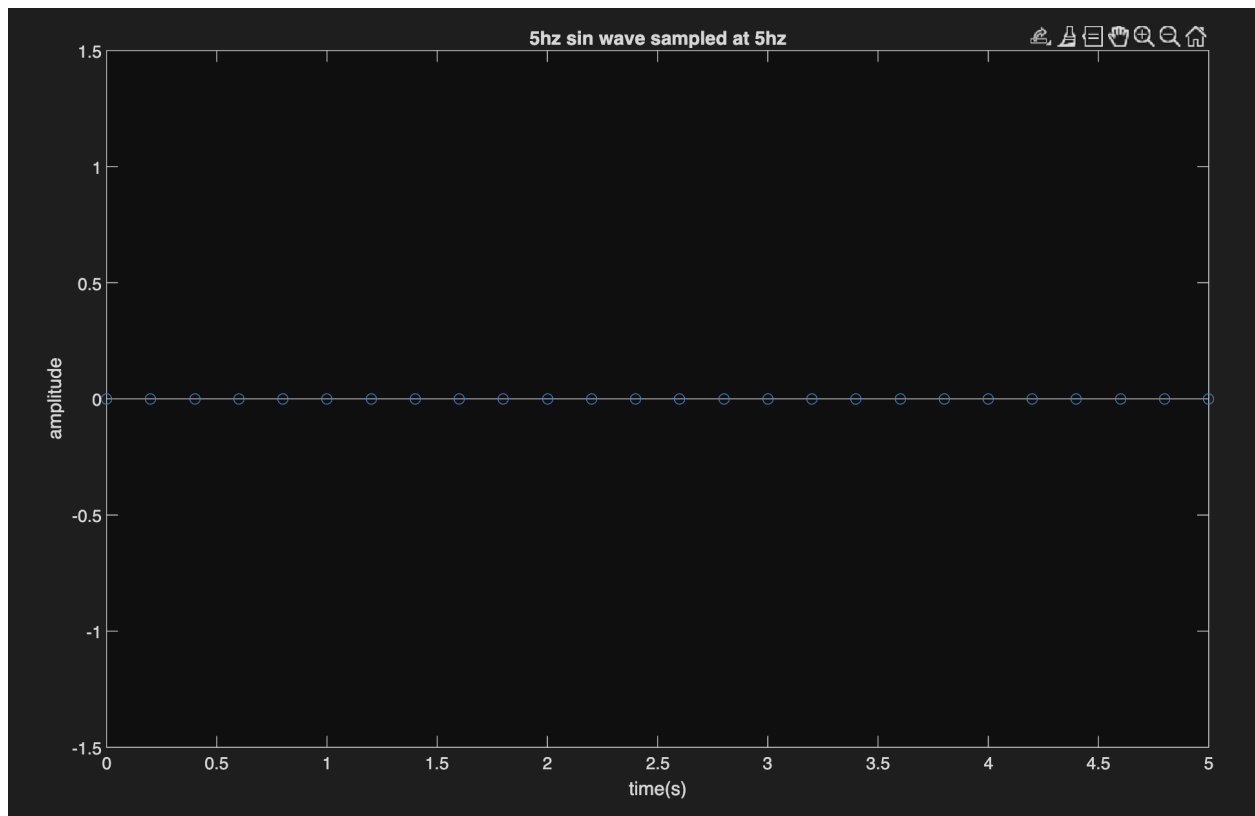
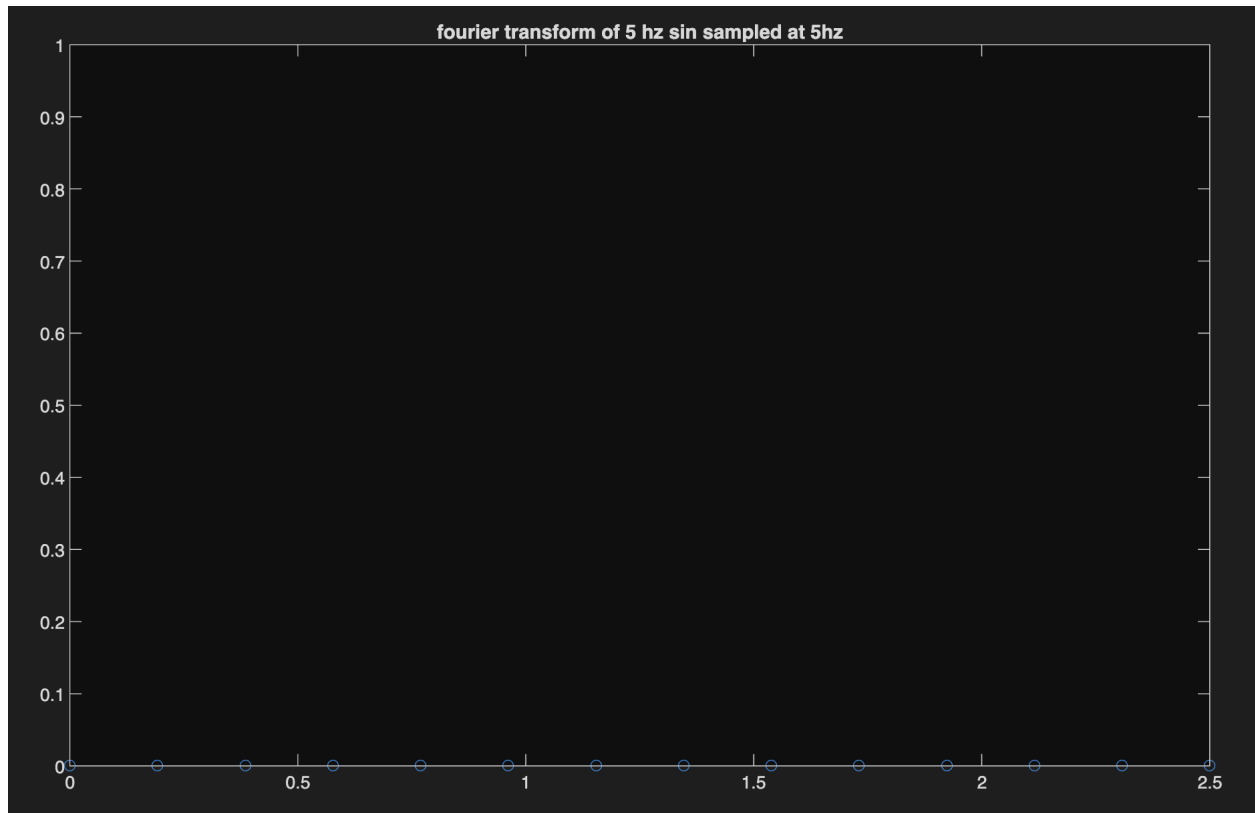


Part 1: Generate a Simple Sine Wave

- Create a sine wave signal:
 - Duration: 5 seconds
 - Sampling Rate: 5 Hz
 - **Frequency: 5 Hz**
- Plot the signal in the time domain.
- Perform frequency analysis using the **Fourier Transform** (e.g., `np.fft.fft` in Python or `fft` in MATLAB).
- Plot the magnitude spectrum and interpret the dominant frequency.





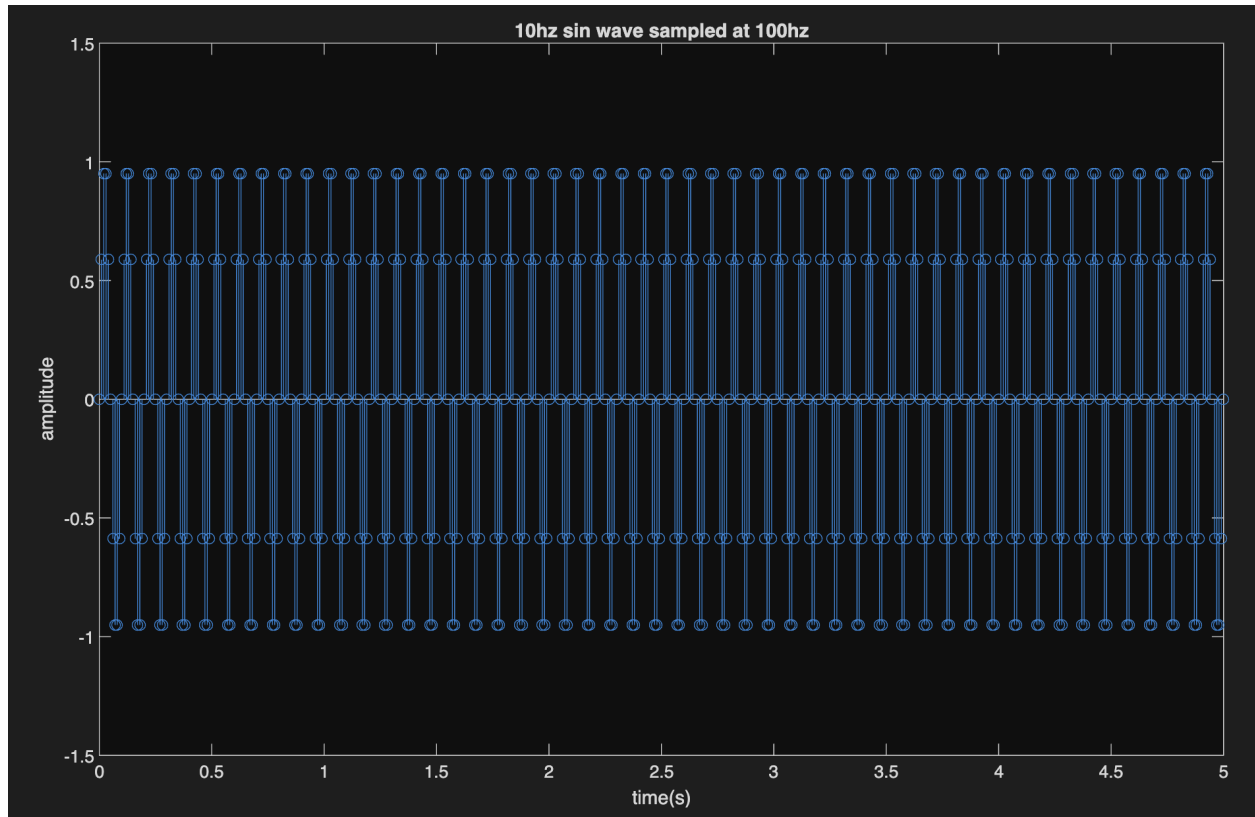
Interpretation: A sine wave sampled at its own frequency will give a constant and no new inferences can be made. Since, no initial phase in $\sin(2\pi ft)$, constant zero was sampled both in time and frequency domain. Nyquist criteria: $f_s \geq 2f_{\max}$.

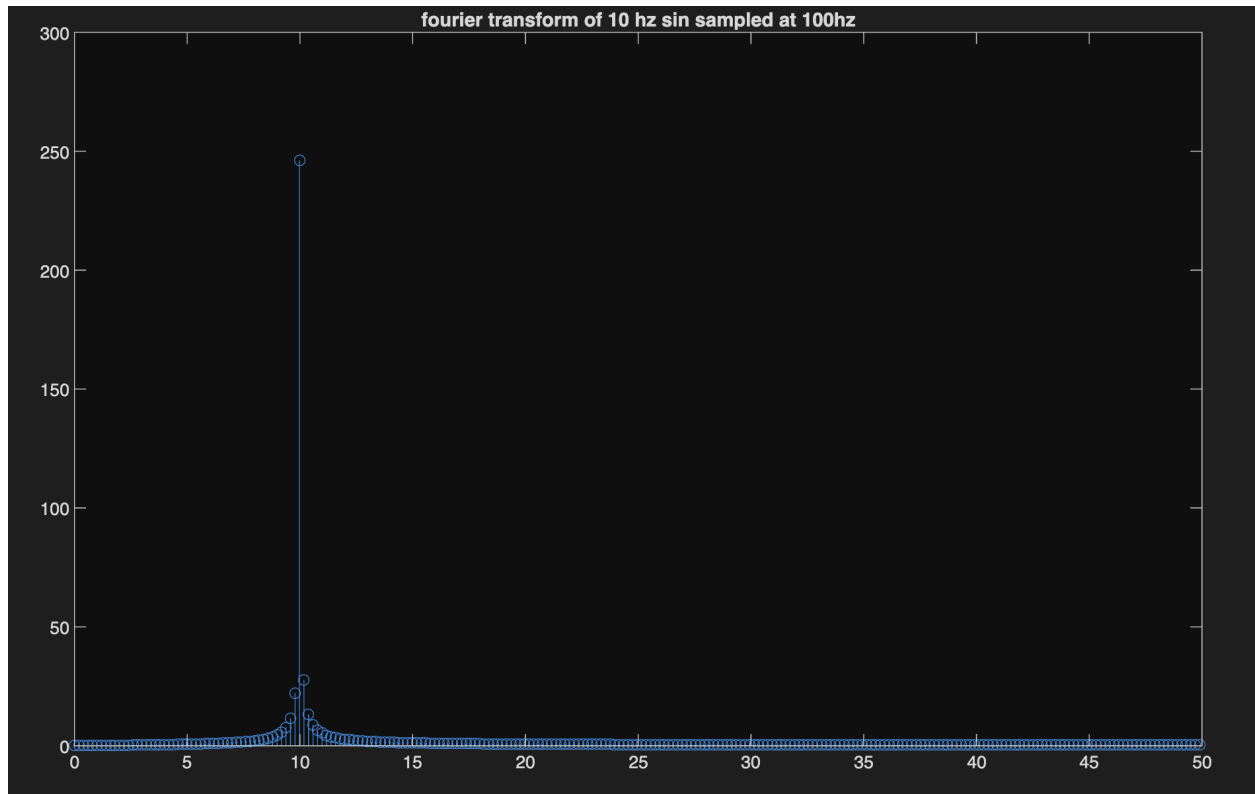
Part 2: Generate Higher Frequency Sine Waves

Repeat the above steps for the following:

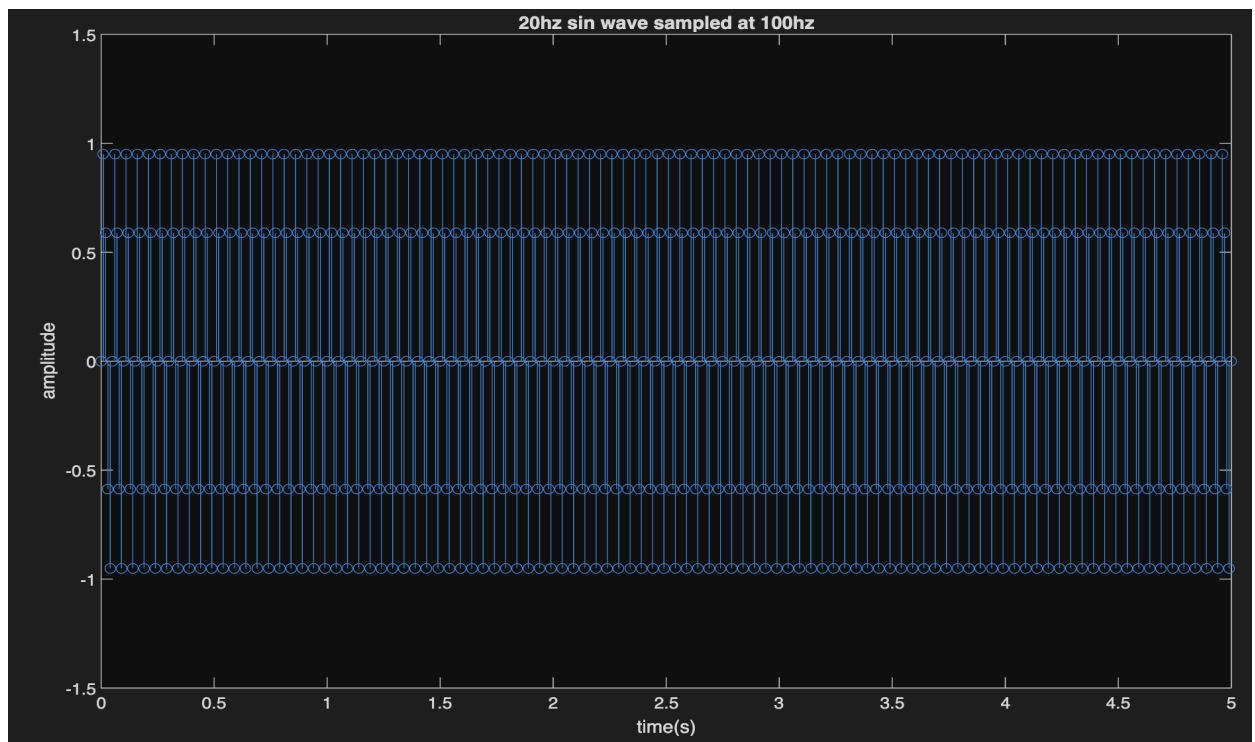
- Frequency: **10 Hz**
- Frequency: **20 Hz**

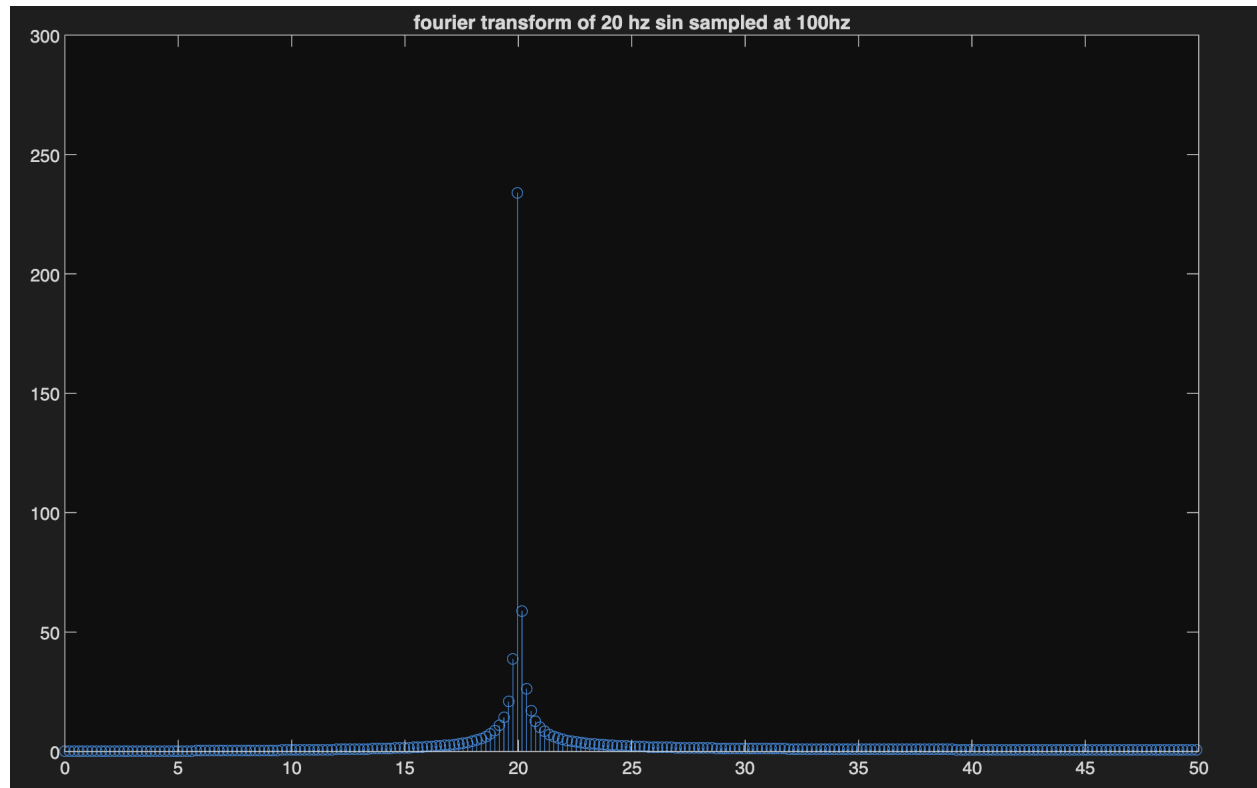
Note: You may increase the sampling rate (e.g., to **100 Hz**) for these signals to meet the Nyquist criterion and avoid aliasing.





Interpretation: no aliasing since $f_s \geq 2f_{\max}$ ($100 \geq 20$). Dominant frequency is 10 hz, given signal was of 10 hz sin wave.

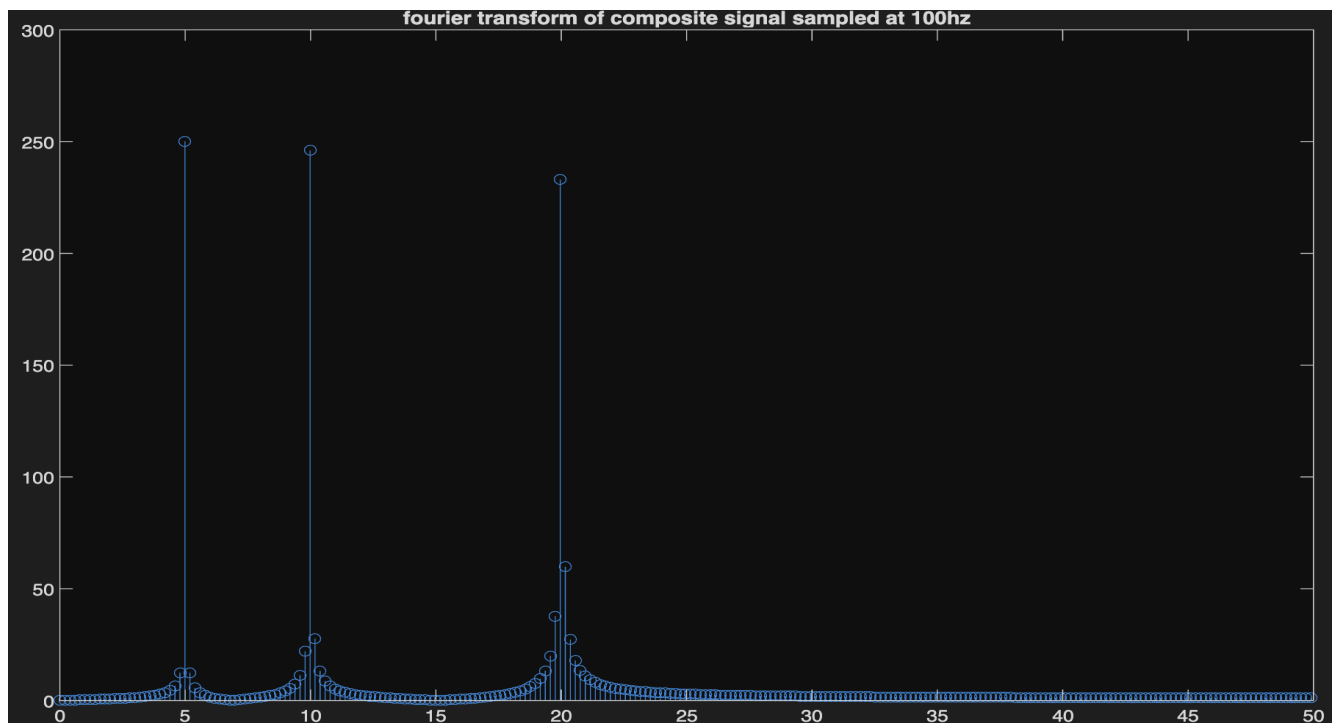
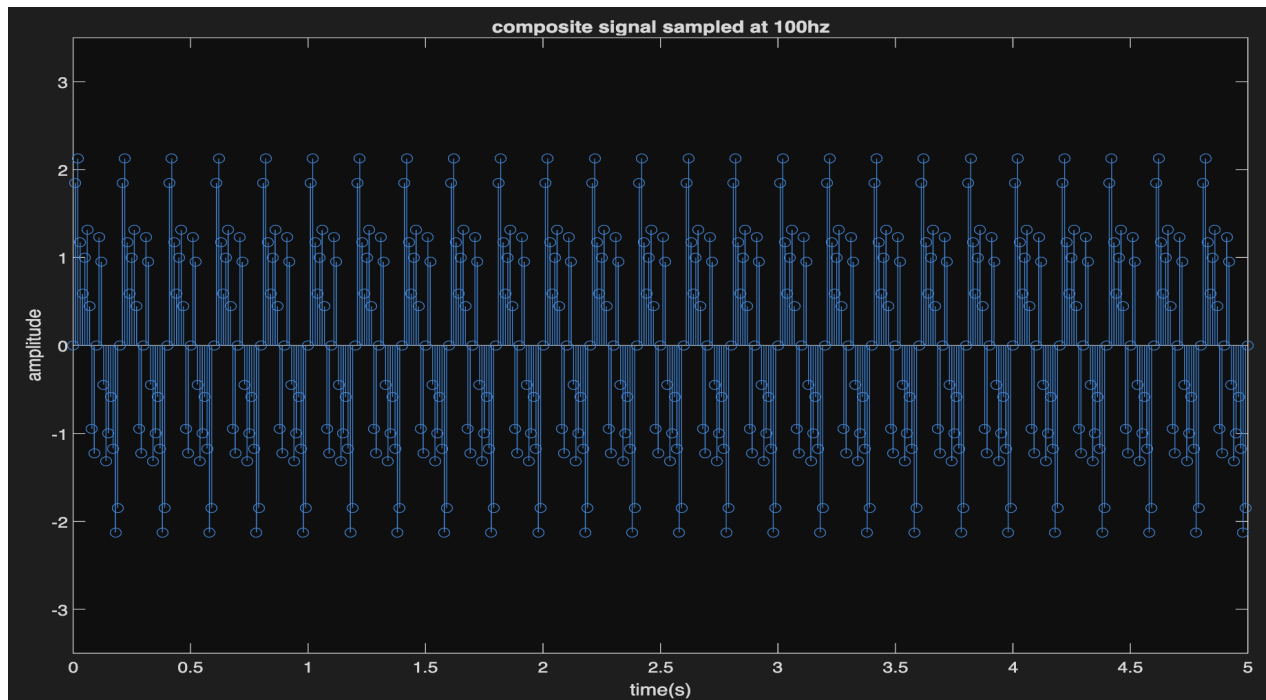




Interpretation: no aliasing since $f_s \geq 2f_{\max}$ ($100 \geq 40$). Dominant frequency is 20 hz, given signal was of 20 hz sin wave.

Part 3: Combine Signals

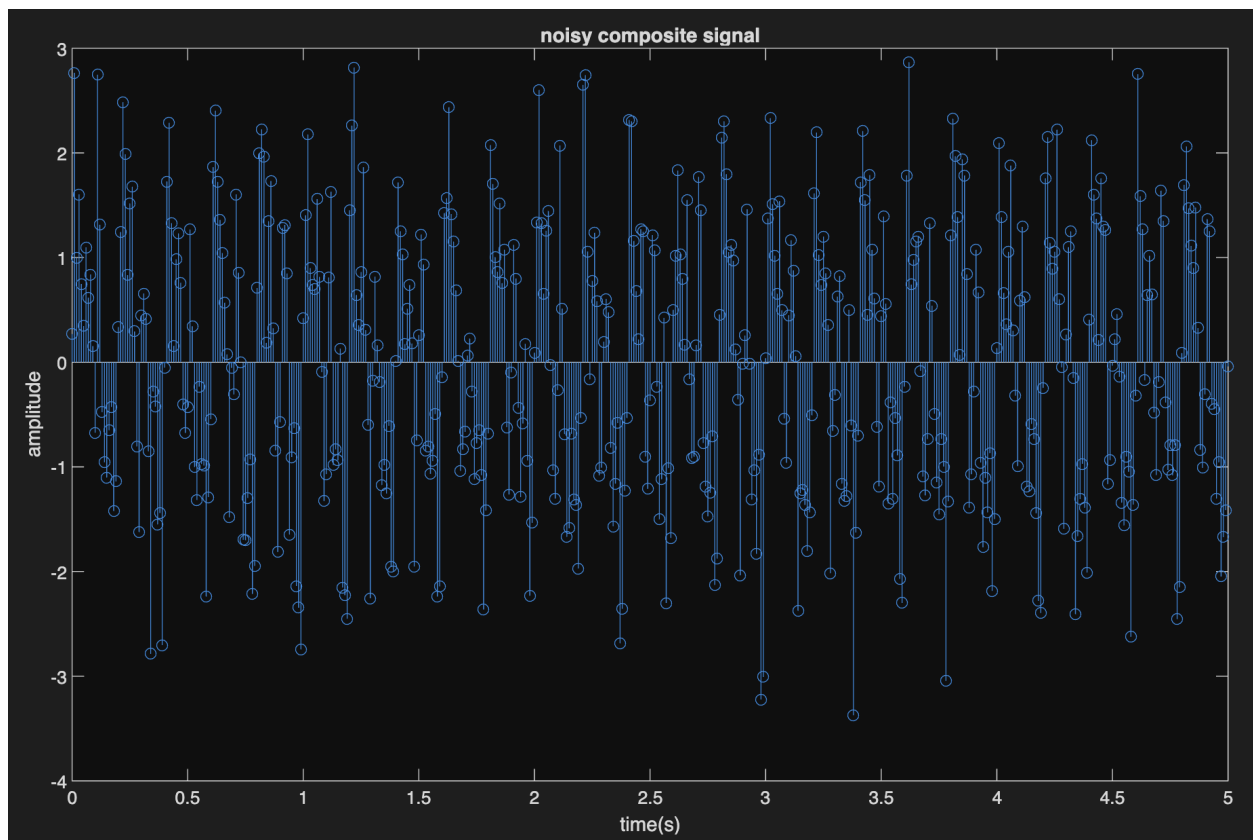
- Combine the three sine waves (5 Hz, 10 Hz, 20 Hz) into a **single composite signal**.
- Plot the time-domain representation of the combined signal.
- Perform Fourier Transform and plot the frequency spectrum.
- Identify the presence of all three frequency components in the frequency domain.

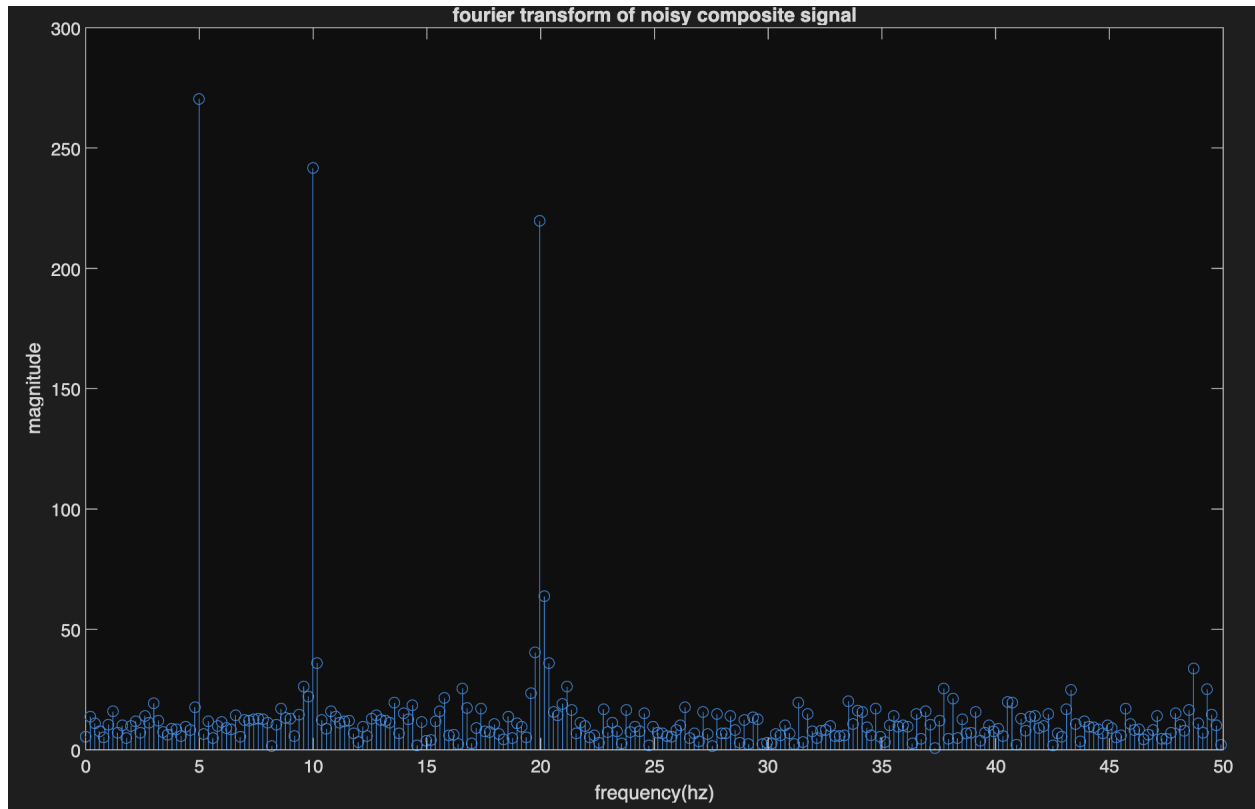


Interpretation: dominant frequencies 5,10 and 20 hz in fourier transform,since signal was made up of 5,10,20 hz sin waves.

Part 4: Add Noise

- Add **random noise** (e.g., Gaussian noise using `np.random.normal`) to the combined signal.
- Plot the noisy signal in the time domain.
- Perform Fourier Transform and plot the frequency spectrum.
- Discuss how noise affects the frequency representation.





Interpretation: adding gaussian noise will not affect dominant frequencies.

Central limit theorem:

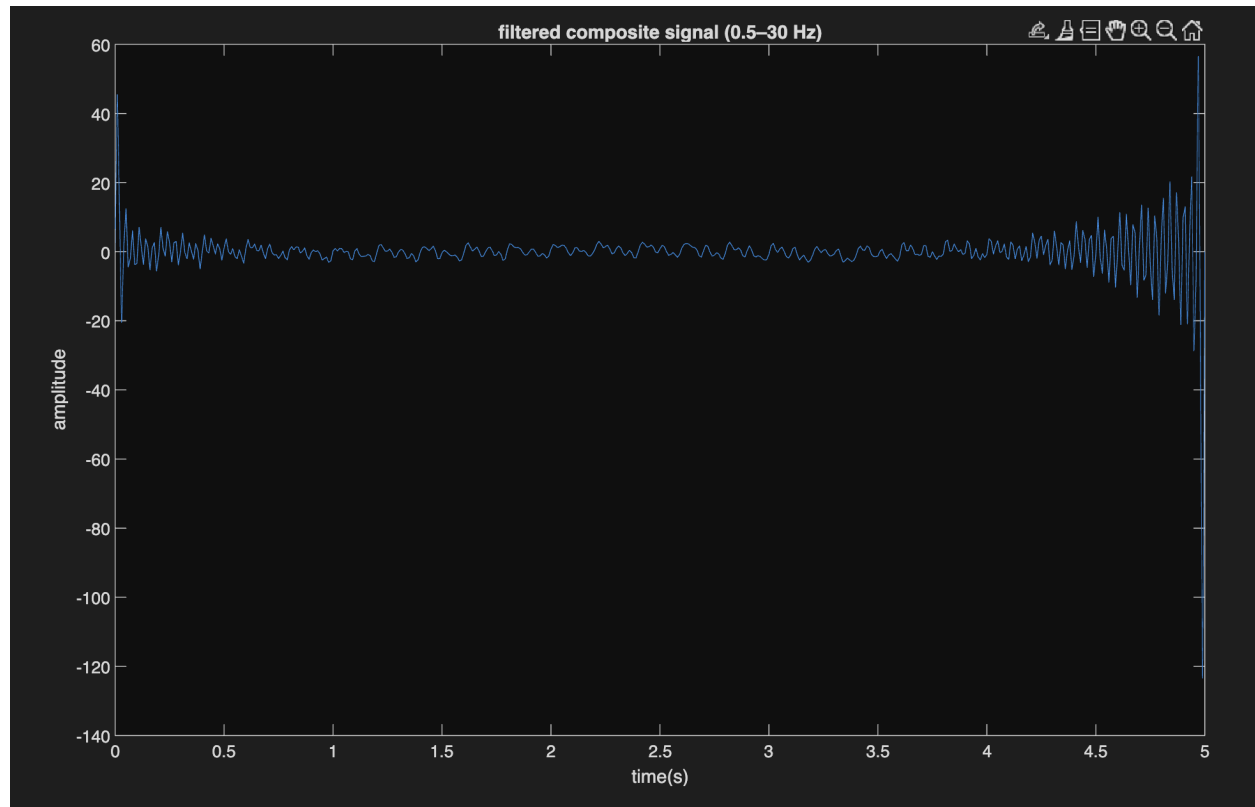
When many tiny random disturbances combine, they ALWAYS become Gaussian.

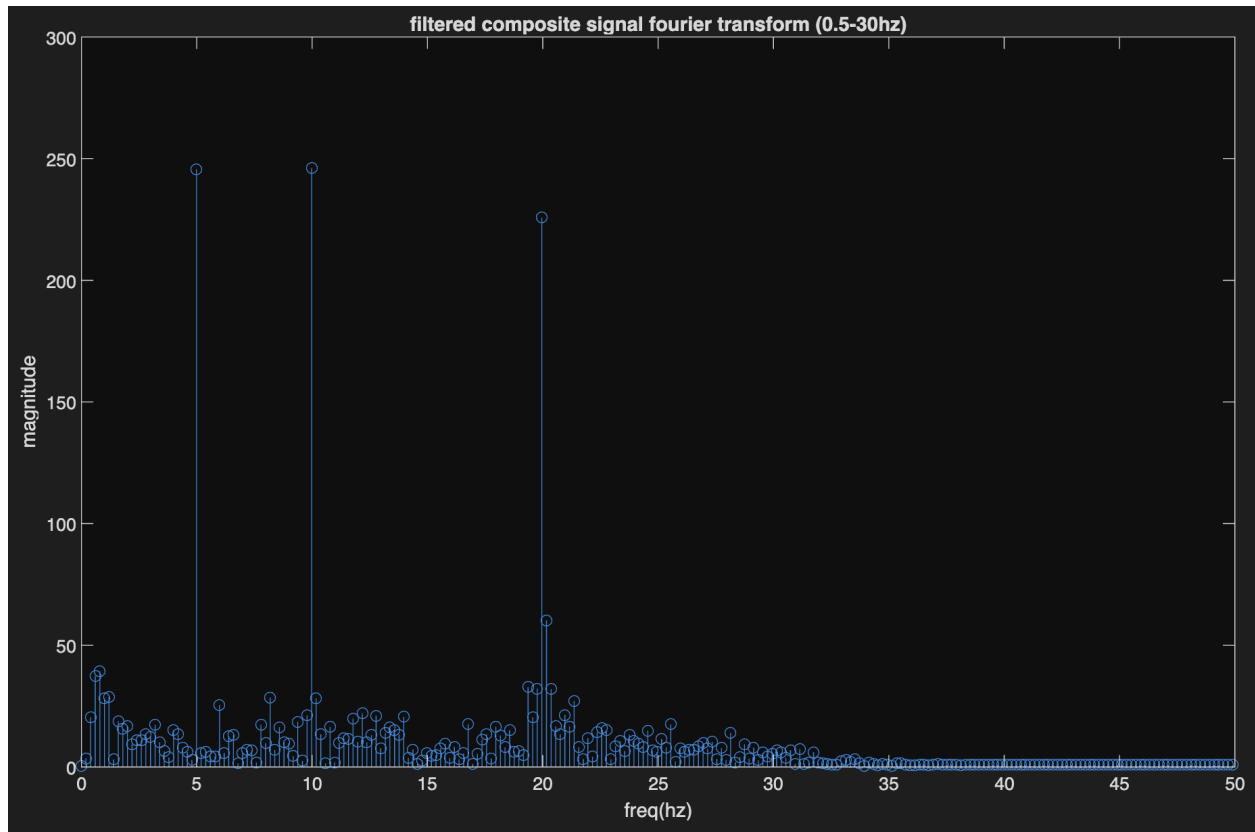
EEG noise = brain activity + muscle + electrode + environment + circuit => **Gaussian**.

Hence the noises dont disturb dominant frequencies.

Part 5: Filtering

- Apply a **bandpass filter** (e.g., 0.5–30 Hz) to the noisy signal and analyze the result.
- Compare the spectrum of the raw noisy signal and the filtered signal.





The bandpass filter allows only frequencies between 0.5 to 30hz to pass and attenuates all other frequencies.

Effect on time domain signal: the signal has been smoothed out. The beginning and end show huge ringing due to filter warm-up and wind-down. The middle portion is the actual filtered signal.

Effect on frequency domain: the signals dominant frequency components can be seen, 5hz, 10hz and 20hz. Frequencies outside the range [0.5,30] have been attenuated.